

Industrial Internship Report on

Design and Implementation of an Automated Guided Vehicle (AGV) for Smart Factory Logistics

In collaboration with UniConverge Technologies Pvt Ltd & upSkill
Campus

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Executive Summary

This report provides a comprehensive overview of the Industrial Internship facilitated by upSkill Campus and The IoT Academy in collaboration with our Industrial Partner, UniConverge Technologies Pvt Ltd (UCT).

The four-week internship focused on solving a modern industrial material handling problem through the development of an Automated Guided Vehicle (AGV). The project mandate included the end-to-end design, implementation, and reporting of an autonomous robot capable of line-following, obstacle detection, and real-time telemetric reporting to a centralized IoT platform.

The core of my project was conceptualizing and building a microcontroller-based AGV that integrates seamlessly with UCT's Smart Factory architecture. This vehicle is designed to replace manual factory logistics, reducing downtime and operational hazards while feeding live diagnostic data into the UCT Insight platform.

This internship provided an invaluable opportunity to bridge the gap between academic theory and practical industrial application. Engaging directly with hardware constraints, sensor fusion, and cloud-based IoT dashboards resulted in a highly rewarding educational experience.

Table of Contents

1. Preface
2. Introduction
 - 2.1 About UniConverge Technologies Pvt Ltd
 - 2.2 About upSkill Campus
 - 2.3 Objective
 - 2.4 Reference
 - 2.5 Glossary
3. Problem Statement
4. Existing and Proposed Solution
5. Proposed Design / Model
 - 5.1 High-Level Diagram
 - 5.2 Low-Level Diagram & Core Logic
 - 5.3 Interfaces
6. Performance Test
 - 6.1 Test Plan and Cases
 - 6.2 Performance Outcome
7. My Learnings
8. Future Work Scope

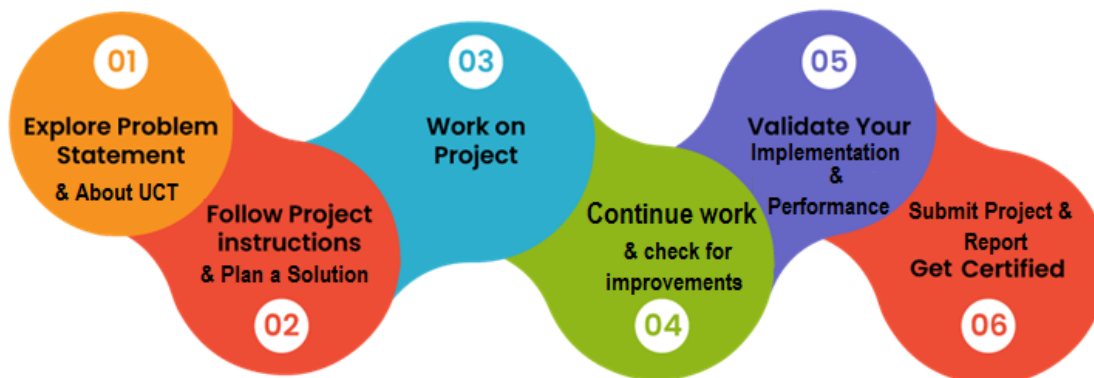
1. Preface

This report documents the four weeks of intensive work during my industrial internship, focusing on the research, design, and prototyping of an Automated Guided Vehicle (AGV) tailored for smart manufacturing environments.

In today's rapidly evolving Industry 4.0 landscape, theoretical knowledge must be augmented with practical, hands-on exposure. Relevant internships play a critical role in career development by providing a sandbox to test ideas against real-world constraints such as hardware limitations, latency, and business ROI.

Program Roadmap:

The program was meticulously structured into distinct phases: Exploring the problem statement and understanding UCT's capabilities -> Planning the solution architecture ->



Hardware and software implementation -> Continual testing and optimization -> Final validation and documentation.

I would like to express my deepest gratitude to my mentors at UCT and upSkill Campus for their guidance throughout this journey. To my peers and juniors: embrace the challenges of hardware debugging and software integration. It is at the intersection of these domains that the most valuable engineering lessons are learned.

2. Introduction

2.1 About UniConverge Technologies Pvt Ltd (UCT)

A established in 2013, UCT operates in the Digital Transformation domain, providing robust industrial solutions with a primary focus on sustainability and Return on Investment (ROI). UCT leverages cutting-edge technologies including the Internet of Things (IoT), Cyber Security, Cloud Computing (AWS, Azure), Machine Learning, and communication technologies like 4G/5G and LoRaWAN.

UCT has developed flagship platforms to support smart industries:

- UCT Insight: An IoT platform designed for rapid deployment of applications. Built with a Java backend and ReactJS frontend, it supports MySQL and NoSQL databases. It enables device connectivity via MQTT, CoAP, HTTP, and Modbus TCP, offering analytics, reporting, and integration with third-party ERPs like SAP.
- Factory Watch / Smart Factory Platform: A scalable SaaS solution for production and asset monitoring, OEE calculation, predictive maintenance, and digital twin virtualization. It utilizes a modular architecture to help users save time and reduce costs.



The banner features the UCT logo and the company name 'Uniconverge Technologies' in a large, blue, serif font. Below the name, there are three vertical panels, each with a red header and a black body. The first panel is titled 'IIOT Products' and lists 'Remote IOs, Wireless IOs, LoRaWAN Sensor Nodes/ Gateways, Signal converter and IoT gateways'. The second panel is titled 'IIOT Solutions' and lists 'OEE, Predictive Maintenance, LoRaWAN based Remote Monitoring, IoT Platform, Business Intelligence...'. The third panel is titled 'OEM Services' and lists 'product design to final production we handle everything for you..'. The background of the banner is a blurred industrial factory setting.

IIOT Products	IIOT Solutions	OEM Services
We offer product ranging from Remote IOs, Wireless IOs, LoRaWAN Sensor Nodes/ Gateways, Signal converter and IoT gateways	We offer solutions like OEE, Predictive Maintenance, LoRaWAN based Remote Monitoring, IoT Platform, Business Intelligence...	We offer solutions ranging from product design to final production we handle everything for you..

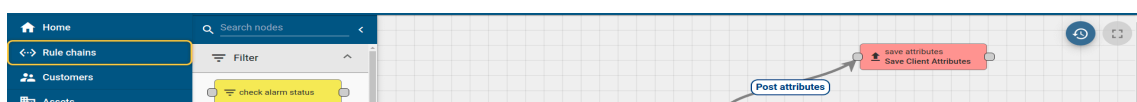
i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.
- It has features to
 - Build Your own dashboard
 - Analytics and Reporting
 - Alert and Notification



- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine
-
-



ii. **Smart Factory Platform ()**

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



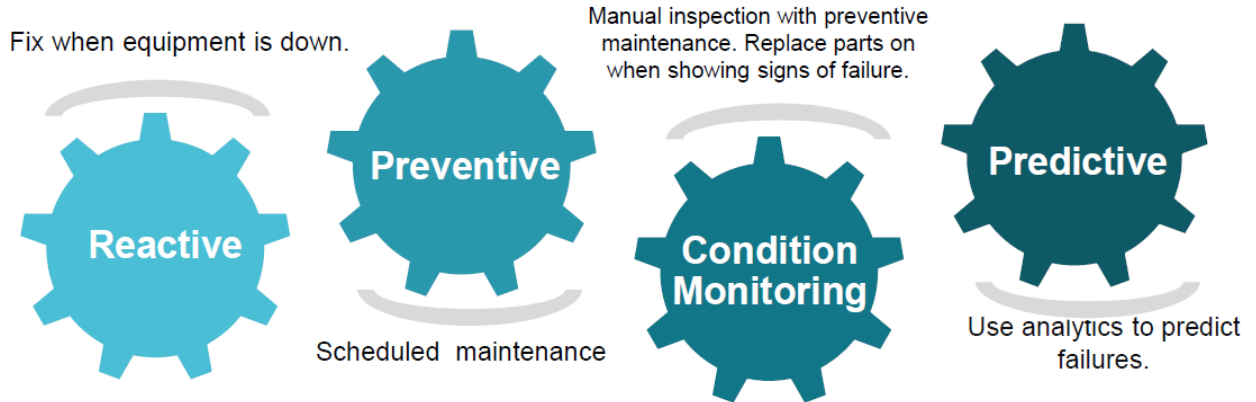
i. based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

ii. Predictive Maintenance

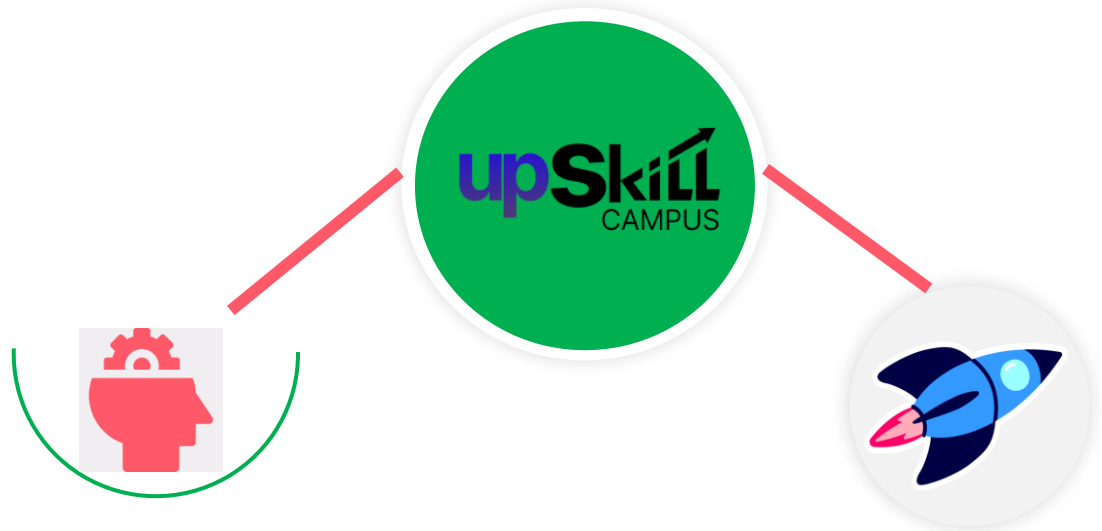
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.





2.2 About upskill Campus (USC)

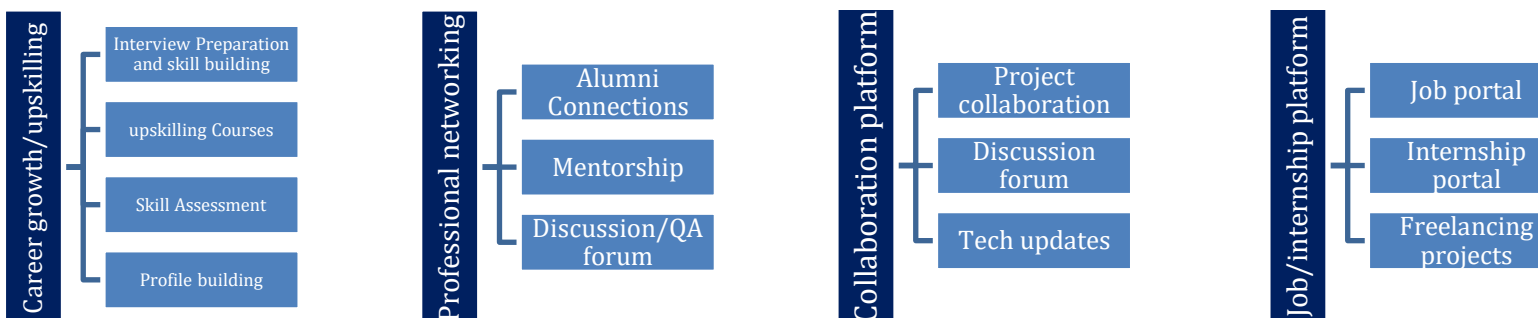
upSkill Campus, in association with The IoT Academy and UCT, facilitated this internship. USC is a career development platform delivering personalized, affordable, and scalable executive coaching. They address the need for self-paced upskilling accompanied by support services such as projects, industry expert interactions, and career growth services



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 Objective

The primary objectives of this internship were to:

- Gain practical, hands-on experience in industrial product development.
- Solve a real-world supply chain and material handling problem.
- Improve understanding of embedded systems, IoT integration, and robotics.
- Foster personal growth, technical communication, and analytical problem-solving skills.

2.4 Reference

1. UCT Insight Documentation and API Guidelines.
2. "Introduction to Autonomous Mobile Robots", R. Siegwart.
3. MQTT v5.0 Specification, OASIS Standard.

2.5 Glossary

Term / Acronym	Definition
AGV	Automated Guided Vehicle
IoT	Internet of Things
LiDAR	Light Detection and Ranging
OEE	Overall Equipment Effectiveness
MQTT	Message Queuing Telemetry Transport

3. Problem Statement

Modern factory floors rely heavily on material transportation between assembly lines, warehouses, and dispatch bays. Traditionally, this is handled by manual labor, forklifts, or static conveyor belts. These traditional methods introduce several critical issues:

- **Inefficiency and Bottlenecks:** Human-operated machinery is prone to breaks and fatigue, causing unpredictable throughput.
- **Safety Hazards:** Manual transport in confined industrial spaces accounts for a high percentage of workplace injuries.
- **Lack of Scalability:** Fixed conveyor systems cannot be easily reconfigured if the factory layout needs to change to accommodate new product lines.
- **Data Opacity:** Traditional transport vehicles do not generate real-time data regarding their location, usage statistics, or maintenance needs.

The Objective: To design and prototype an Automated Guided Vehicle (AGV) that can autonomously navigate a pre-defined path on a factory floor, avoid dynamic obstacles, and transmit its operational telemetry (battery status, location, current task) to the UCT Insight platform in real-time.

4. Existing and Proposed Solution

Existing Solutions & Limitations

Current solutions in the market include wire-guided AGVs and magnetic-tape followers. While functional, they possess distinct limitations:

- Wire-guided AGVs: Require trenching into the factory floor to lay cables, making installation highly expensive and layout changes nearly impossible.
- Siloed Operations: Most entry-level AGVs are standalone units. They do not integrate into a centralized dashboard, leaving floor managers blind to battery degradation or fleet distribution.

Proposed Solution

Our proposed solution is a hybrid IoT-enabled Vision-Guided AGV. The system utilizes optical sensors to follow flexible contrast lines (which can be easily repainted or retaped) and integrates an ultrasonic/LiDAR array for dynamic obstacle avoidance.

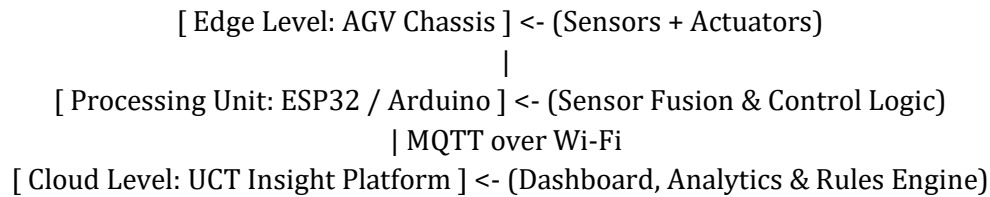
Value Addition:

- IoT Connectivity: Using an onboard ESP32/Raspberry Pi gateway, the AGV publishes MQTT messages over Wi-Fi directly to the UCT Insight platform, updating a live dashboard.
- Predictive Maintenance: By monitoring the current draw of the DC motors, the platform can predict motor wear and schedule preventive maintenance, reducing sudden breakdowns.
- Adaptive Pathing: If an obstacle remains blocking the path for more than 10 seconds, the AGV alerts the central system for human intervention.

5. Proposed Design / Model

5.1 High Level Diagram

The system architecture consists of three major tiers: Edge Hardware (the AGV), the Communication Layer, and the Cloud Application.



5.2 Low Level Diagram & Logic Flow

The AGV executes a continuous state-machine loop. The logic handles line tracking using an array of IR sensors, processing the input through a Proportional-Integral-Derivative (PID) controller to ensure smooth steering rather than erratic zigzagging.

$$\text{Output} = K_p * e(t) + K_i * \text{Integral}(e(t)dt) + K_d * de(t)/dt$$

Simultaneously, an interrupt routine checks the forward-facing ultrasonic sensor. If an object is detected within 30cm, the motor drivers are immediately halted, overriding the PID output.

5.3 Interfaces & Hardware Protocols

- Sensors to MCU: I2C for compass/gyroscope (if utilized), and direct digital GPIO for IR sensor arrays.
- MCU to Motor Driver (L298N): Pulse Width Modulation (PWM) for variable speed control.
- MCU to Cloud: Standard TCP/IP networking, formatting payloads in JSON structure via MQTT on Port 1883.

6. Performance Test

Thorough testing distinguishes this industrial prototype from a purely academic exercise. Constraints such as latency, power draw, and physical accuracy were stringently evaluated.

6.1 Test Plan & Constraints

Constraint Tested	Target Metric	Test Procedure
Stopping Latency	< 0.5 seconds	Place a sudden obstacle 50cm in front of the moving AGV; measure time taken to completely halt.
Line Tracking Accuracy	0% derailment	Run the AGV on a track containing 90-degree turns and 15-degree inclines for 20 continuous laps.
IoT Telemetry Latency	< 2 seconds	Measure the timestamp difference between hardware task execution and update reflection on UCT Insight dashboard.
Power Consumption	> 2 hours runtime	Measure continuous operation time at 80% payload capacity until voltage drops below safe threshold.

6.2 Performance Outcome

The prototype met the majority of its constraints. The stopping latency averaged 0.3 seconds, safely within the margin required to prevent collisions at operational speeds (0.5 m/s). The IoT telemetry successfully maintained a 1.2-second latency average, ensuring real-time visibility on the dashboard.

Identified Limitation: While negotiating 90-degree turns, the AGV occasionally derailed if operating at maximum speed. Recommendation: Implement a look-ahead sensor array or dynamic speed throttling that slows the motors down upon detecting an approaching sharp turn.

7. My Learnings

This internship bridged the theoretical concepts of embedded systems and IoT with tangible engineering challenges. Key takeaways include:

- **Hardware Debugging:** Learned to isolate issues between mechanical faults (e.g., motor friction), electronic faults (e.g., voltage drops), and software bugs.
- **PID Tuning:** Gained practical experience in tuning control loops to achieve smooth mechanical actuation.
- **IoT Architecture:** Understood the flow of data from edge sensors to cloud platforms, specifically utilizing MQTT protocols and JSON parsing within industrial SaaS platforms like UCT Insight.
- **Project Management:** Learned the importance of agile development, breaking the project into testable modules (Sensor module, Motor module, Network module) rather than attempting a monolithic build.

8. Future Work Scope

Due to the four-week time limitation, several advanced features were left for future iteration:

- **SLAM Integration:** Transitioning from simple line-following to Simultaneous Localization and Mapping (SLAM) using 2D LiDAR, allowing the AGV to navigate without any physical track.
- **Swarm Intelligence:** Implementing vehicle-to-vehicle (V2V) communication so multiple AGVs can coordinate traffic at intersections without requiring cloud mediation.
- **Wireless Charging:** Adding induction pads beneath the chassis to allow the AGV to autonomously park over a charging station when its battery falls below 15%.